BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, DG KHAN

INTERMEDIATE (12TH CLASS) EXAMINATION, 2024

CHEMISTRY

Past Paper 2024 • Paper Code: 8485 / 8488

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

OBJECTIVE

Time Allowed: 20 MINUTES | **Maximum Marks:** 17 | **Paper Code:** 8485

Note: You have four choices for each objective type question as A , B , C and D . The choice which you think is correct, fill that circle in front of that question number. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question.

Q1. Planar triangular shape is associated with which set of hybrid orbitals

(A) sp^3
(B) sp
(C) sp^2
(D) dsp^2

Q2. Formula of chloroform is

(A) CH_3Cl
(B) CCl_4
(C) CH_2Cl_2
(D) $CHCl_3$

Q3. During nitration of benzene , the active nitrating agent is

(A) NO_3
(B) NO_2^+
(C) NO_2
(D) HNO_3

Q4. When Ethanol ($CH_3 – CH_2 – OH$) is made to react with ethyl magnesium bromide the product formed is

(A) CH_4
(B) $CH_2 = CH_2$
(C) C_2H_2
(D) $CH_3 – CH_3$

Q5. Which compound is called a universal solvent ?

(A) H_2O
(B) CH_3OH
(C) C_2H_5OH
(D) $CH_3 – O – CH_3$

Q6. One of the following compounds will react with Tollen's reagent.

(A) $CH_3 - C(=O) - H$
(B) $CH_3 - C(=O) - CH_3$
(C) $CH_3 - C(=O) - OH$
(D) $CH_3 - C(=O) - CH_2 - CH_3$

Q7. One of the following is not a fatty acid

(A) Acetic acid
(B) Propanoic acid
(C) Butanoic acid
(D) Phthalic acid

Q8. Which one of the following enzymes brings about the hydrolysis of fats ?

(A) Urease
(B) Maltase
(C) Lipase
(D) Zymase

Q9. Which is not a calcarious material ?

(A) Lime
(B) Clay
(C) Marble
(D) Marine Shell

Q10. Disinfection of chlorine is

(A) Inexpensive
(B) Expensive
(C) Rapid
(D) Slow

Q11. Ecosystem is a smaller unit of

(A) Lithosphere
(B) Hydrosphere
(C) Atmosphere
(D) Biosphere

Q12. Select the two normal elements are present in Fifth period

(A) Rb , Sr
(B) Cs , Ba
(C) Fr , Ra
(D) La , Hf

Q13. The mineral ($CaSO_4 . 2H_2O$) has the general name

(A) Gypsum
(B) Dolomite
(C) Calcite
(D) Epsom salt

Q14. One of the element forms an ion with charge 3+

(A) Beryllium
(B) Aluminium
(C) Carbon
(D) Silicon

Q15. Choose the gas which is obtained by the reaction of Formic acid with conc: H_2SO_4

(A) C_2H_2
(B) C_2H_4
(C) CO_2
(D) CO

Q16. One of the halogen occurs naturally in a positive oxidation state

(A) Fluorine (B) Chlorine

(C) Iodine (D) Bromine

Q17. Group VI – B of transition elements contains

(A) Zn , Cd , Hg (B) Cr , Mo , W

(C) Fe , Ru , Os (D) Mn , Te , Re

SUBJECTIVE PART SECTION – I / SECTION-II

Time Allowed: 2 HRS 40 MINUTES | **Maximum Marks:** 68

Q2. (Marks 16) Write short answers any Eight (8) of the following

i Why alkali metals are very reactive ?

ii What are the s – block elements and why are they called so ?

iii How do transition elements display colours ?

iv K_2CrO_4 and $K_2Cr_2O_7$ show similar properties. Justify this.

v How does brown pulp converted into white pulp in paper industry ?

vi Give formula of (i) Dolomite (ii) Natron

vii What will happen when Lithium Hydride is treated with water.

viii Starting from (primary) alkyl halide prepare ethylamine

ix How the molecular mass of polymer is determined ?

x What is starch ? Write its two uses.

xi What are lipids ? What is their importance ?

xii How Grignard reagent react with CH_3CHO ?

Q3. (Marks 16) Write short answers any Eight (8) of the following

i Why is SO_3 dissolved in H_2SO_4 and not in H_2O ?

ii Write down any four uses of HNO_3

iii What are Freons and Teflon ?

iv Write down any four applications of noble gases.

v What are alicyclic compounds ? Give its two examples.

vi Define knocking. How does it improve ?

vii How formaldehyde is prepared from ethene ?

viii How acetylene is prepared on industrial scale ?

ix Why alkanes are less reactive ?

x Differentiate between oxidizing and reducing smog.

xi How chlorofluorocarbons destroy ozone ?

xii What is dissolved oxygen ?

Q4. (Marks 12) Write short answers any Six (6) of the following

i What is the Chemistry of Borax – bead Test ?

ii Write down any two uses of Aluminium.

iii Why are liquid silicones preferred over ordinary organic lubricants ?

iv What are the major products of bromination of the following compounds? (a) Toluene (b) Benzaldehyde

v Ethanol gives different products with conc H_2SO_4 under different conditions , give reactions.

vi Write IUPAC names of the following compounds. (a) $(CH_3)_2$ CH – OH (b) $(CH_3)_2$ CH CH_2 OH

vii Why formaldehyde does not give aldol condensation reaction ?

viii What is internal salt of amino acids ?

ix Write down the reaction of acetic acid with ammonia and its product on heating.

> Note: SECTION-II — Note: Attempt any Three questions from this section 8 x 3 = 24

Q5.

(A) Discuss the position of hydrogen in a periodic table with IV A group elements.

(B) How is sodium prepared commercially by Down's Cell ?

Q6.

(A) Write down four uses of Chlorine in daily life.

(B) Explain preparation of urea fertilizer on large scale.

Q7.

(A) Define atomic orbital hybridization. Explain sp hybridization with formation of ethyne.

(B) Explain the mechanism of SN_1 reactions.

Q8.

(A) How will you prepare from ethyne. (i) Acetaldehyde (ii) Oxalic acid (iii) Benzene (iv) chloroprene

(B) How will you prepare from Acetaldehyde (i) Lactic acid (ii) Acetal

Q9.

(A) Discuss the nitration of benzene with reaction mechanism.

(B) How will you identify 1 – propanol , 2 – propanol and 2 – methyl – 2 – propanol ? Justify the answer with chemical test.

GROUP – II

OBJECTIVE

Maximum Marks: 17 | **Paper Code:** 8488

Note: You have four choices for each objective type question as A , B , C and D . The choice which you think is correct, fill that circle in front of that question number. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question.

Q1. Which of the following is not a fatty acid ?

(A) Propanoic acid
(B) Acetic acid
(C) Phthalic acid
(D) Butanoic acid

Q2. A polymeric substance that is formed in a liquid state and then hardened to a rigid solid is called a

(A) Fibre
(B) Plastic
(C) Varnish
(D) Polyamide resins

Q3. Which is not a calcarious material ?

(A) Lime
(B) Clay
(C) Marble
(D) Marine shell

Q4. The main pollutant of leather tanneries in the waste water is due to the salt of

(A) Lead
(B) Chromium (VI)
(C) Copper
(D) Chromium (III)

Q5. The capacity of organic matter in natural water to consume oxygen within a period of five days is called

(A) DO
(B) BOD
(C) COD
(D) PAN

Q6. Coinage metals are

(A) Ni , Pd , Pt
(B) Cu , Ag , Au
(C) As , Al , Pd
(D) Fe , Si , Sn

Q7. Which element is deposited at cathode during the electrolysis of brine in diaphragm cell ?

(A) H_2
(B) Na
(C) Cl_2
(D) O_2

Q8. The chief ore of Aluminium is

(A) $Na_3\ AlF_6$
(B) $Al_2O_3 . 2H_2O$
(C) Al_2O_3
(D) $Al_2O_3 . H_2O$

Q9. Nitrous acid reacts with aminobenzene to produce

(A) Urea
(B) Phenol
(C) Toluene
(D) NH_3

Q10. Which of the following represents the correct electronic configuration of the outermost energy level of an element of zero (VIII A) group in the ground state ?

(A) s^2p^2
(B) s^2p^4
(C) s^2p^5
(D) s^2p^6

Q11. Coordination number of Pt in [Pt Cl (NO_2) $(NH_3)_4$] is

(A) 2
(B) 4
(C) 1
(D) 6

Q12. Select from the following the one which is alcohol

(A) $CH_3 – CH_2 – OH$
(B) $CH_3 – O – CH_3$
(C) $CH_3\ COOH$
(D) $CH_3 – CH_2 – Br$

Q13. The formula of chloroform is

(A) CH_3Cl
(B) CCl_4
(C) CH_2Cl_2
(D) $CHCl_3$

Q14. During nitration of benzene , the active nitrating agent is

(A) NO_2
(B) NO_2^+
(C) NO_2^-
(D) HNO_3

Q15. Which reagent does not produce ethane by reacting with ethyl magnesium chloride in the presence of dry ether ?

(A) H_2O
(B) NH_3
(C) HCHO
(D) C_2H_5OH

Q16. Which compound will have maximum repulsion with water ?

(A) C_6H_6 (B) C_2H_5OH

(C) CH_3OCH_3 (D) CH_3CH_2OH

Q17. Which of the following reagents will react with both aldehydes and ketones ?

(A) Grignard reagent (B) Fehling's reagent

(C) Benedict's reagent (D) Tollen's reagent

SUBJECTIVE PART SECTION – I / SECTION-II

Time Allowed: 2 HRS 40 MINUTES | **Maximum Marks:** 68

Q2. (Marks 16) Write short answers any Eight (8) of the following

i Why diamond is a non – conductor but graphite is a fairly good conductor ?

ii Define ionization energy. How does it vary in the periodic table ?

iii Why 2% gypsum is added to Cement ?

iv Give the formula of (i) Asbestos (ii) Soapstone (Talc)

v How does the process of galvanizing or Zinc Coating protect the iron from rusting ?

vi How is the chromate ions converted into dichromate ions. Give the reaction involved.

vii Give the two factors which govern the reactivity of alkyl halides.

viii How propanoic acid is prepared from Ethyl magnesium Bromide ?

ix What are thermoplastic polymers. Give two examples.

x Define polysaccharides. Give two examples

xi How the proteins are denatured ?

xii Define lignin. Why is it removed from pulp ?

Q3. (Marks 16) Write short answers any Eight (8) of the following

i Why the elements of Group VIA other than oxygen shows more than two oxidation states ?

ii What is meant by Fuming Nitric acid ?

iii Arrange F^- , Cl^- , I^- , Br^- in order of decreasing size.

iv Name the Halogen used in water treatment.

v Draw flow sheet diagram for the formation of Anthracite.

vi Differentiate between petroleum and crude oil.

vii How will you synthesize Acetaldehyde from C_2H_2 ?

viii Compare the reactivity of ethane and ethene.

ix Describe polymerization of ethene.

x Define Biochemical Oxygen Demand (BOD) for the quality of water.

xi Describe how pesticides are dangerous to humans ?

xii Describe Reprocessing for the Recycling of Plastics.

Q4. (Marks 12) Write short answers any Six (6) of the following

i What are semiconductors and name elements and compounds act as semiconductors ?

ii Give four uses of Boric Acid.

iii Justify that Boric Acid is monobasic Acid.

iv What objections were raised on Kekule's structure for benzene molecule ?

v How phenol is converted into Bakelite ?

vi How Picric Acid is obtained from phenol ?

vii What is Silver Mirror Test ?

viii How would you convert acetic acid into acetic anhydride ?

ix Discuss strecker synthesis for the preparation of amino acid.

> Note: SECTION-II — Note: Attempt any Three questions from this section 8 x 3 = 24

Q5.

(A) What are hydrides ? Name their types. Give properties of ionic hydrides .

(B) What are general trends of oxides and hydroxides of alkali and alkaline earth metals ?

Q6.

(A) What happens when bleaching powder reacts with (i) HCl (ii) NH_3 (iii) H_2SO_4 (Excess) (iv) CO_2

(B) Describe screening and bleaching steps in Neutral sulphite semi chemical process.

Q7.

(A) Define sp^3 hybridization. Discuss it with a suitable example along with labeled diagram.

(B) Define nucleophilic substitution reaction. Explain SN_2 mechanism in detail.

Q8.

(A) Discuss oxidation reactions of alkynes. (any two)

(B) Describe disproportionation reaction in benzaldehyde , when it is treated with aqueous solution of 50 % NaOH at room temperature.

Q9.

(A) Give preparation of benzene with any four methods.

(B) Give reactions for the preparation of phenyl acetate , benzene cyclohexanol and picric acid from phenol.